1.57 Binomial Hypothesis Testing

Worked Solutions

1)

The significance level is the probability of rejecting the null hypothesis when in fact it is true.

2 a)

$\mathrm{P}(X \leq 10) = 0.0480 < 0.05$

Reject H_0

2 b)

$\mathrm{P}(X \leq 5) = 0.0766 > 0.05$

Fail to reject H_0

2 c)

$\mathrm{P}(X \leq 12) = 0.0743 < 0.1$

Reject H_0

2 d)

$\mathrm{P}(X \leq 40) = 0.0045 < 0.01$

Reject H_0

3 a)

$\mathrm{P}(X \geq 9) = 0.0950 > 0.05$

Fail to reject H_0

3 b)

$\mathrm{P}(X \geq 41) = 0.1748 > 0.05$

Fail to reject H_0

3 c)

$\mathrm{P}(X \geq 26) = 0.0263 < 0.1$

Reject H_0

3 d)

$\mathrm{P}(X \geq 18) = 0.0199 > 0.01$

Fail to reject H_0

4 a)

$25 \times 0.8 = 20$

$\mathrm{P}(X = 15) = 0.01177695$

$\mathrm{P}(X \leq 15) = 0.01733187$

$\mathrm{P}(X \geq 25) = 0.003777893$

p-value $= 0.01733 \ldots + 0.00377 \ldots = 0.0211 < 0.05$

Reject H_0

4 b)

$33 \times 0.72 = 23.76$

$\mathrm{P}(X = 29) = 0.01833201$

$\mathrm{P}(X \geq 29) = 0.0264523$

$\mathrm{P}(X \leq 18) = 0.02417253$

p-value $= 0.0264523 \ldots + 0.02417253 \ldots = 0.0506 > 0.05$

Fail to reject H_0

4 c)

$67 \times 0.38 = 25.46$

$\mathrm{P}(X = 33) = 0.01687621$

$\mathrm{P}(X \geq 33) = 0.03956265$

$\mathrm{P}(X \leq 17) = 0.02044554$

p-value $= 0.03956265 \ldots + 0.02044554 \ldots = 0.0600 < 0.1$

Reject H_0

4 d)

$180 \times 0.05 = 9$

$\mathrm{P}(X = 3) = 0.01362644$

$\mathrm{P}(X \leq 3) = 0.01901408$

$\mathrm{P}(X \geq 16) = 0.01914646$

p-value $= 0.01901408 \ldots + 0.01914646 \ldots = 0.0382 > 0.01$

Fail to reject H_0

5 a)

The critical region is the range of values of X that will reject the null hypothesis.

5 b)

The acceptance region is the range of values of X that will fail to reject the null hypothesis.

6 a)

$\mathrm{P}(X \leq 5) = 0.0481 < 0.05$

$\mathrm{P}(X \leq 6) = 0.1189 > 0.05$

CR is $\{0 \leq X \leq 5\}$

Actual Significance Level $= 4.81\%$

6 b)

$\mathrm{P}(X \leq 15) = 0.0394 < 0.05$

$\mathrm{P}(X \leq 16) = 0.0760 > 0.05$

CR is $\{0 \leq X \leq 15\}$

Actual Significance Level = 3.94%

6 c)

$\mathrm{P}(X \leq 6) = 0.0738 < 0.1$

$\mathrm{P}(X \leq 7) = 0.1507 > 0.1$

CR is $\{0 \leq X \leq 6\}$

Actual Significance Level = 7.38%

6 d)

$\mathrm{P}(X \leq 48) = 0.0070 < 0.01$

$\mathrm{P}(X \leq 49) = 0.0157 > 0.01$

CR is $\{0 \leq X \leq 48\}$

Actual Significance Level = 0.70%

7 a)

$\mathrm{P}(X \geq 12) = 0.0875 > 0.05$

$\mathrm{P}(X \geq 13) = 0.0432 < 0.05$

CR is $\{13 \leq X \leq 40\}$

Actual Significance Level = 4.32%

7 b)

$\mathrm{P}(X \geq 22) = 0.0551 > 0.05$

$\mathrm{P}(X \geq 23) = 0.0295 < 0.05$

CR is $\{23 \leq X \leq 49\}$

Actual Significance Level = 2.95%

7 c)

$\mathrm{P}(X \geq 10) = 0.1733 > 0.1$

$\mathrm{P}(X \geq 11) = 0.0969 < 0.1$

CR is $\{11 \leq X \leq 49\}$

Actual Significance Level = 9.69%

7 d)

$\mathrm{P}(X \geq 104) = 0.01004 > 0.01$

$\mathrm{P}(X \geq 105) = 0.0054 < 0.01$

CR is $\{105 \leq X \leq 125\}$

Actual Significance Level = 0.54%

8 a)

$\mathrm{P}(X \leq 16) = 0.0221 < 0.025$

$\mathrm{P}(X \leq 17) = 0.0413 > 0.025$

$\mathrm{P}(X \geq 32) = 0.02503 > 0.025$

$\mathrm{P}(X \geq 33) = 0.0133 < 0.025$

CR is $\{0 \leq X \leq 16\} \cup \{33 \leq X \leq 60\}$

Actual Significance Level = 3.54%

8 b)

$\mathrm{P}(X \leq 34) = 0.0103 < 0.025$

$\mathrm{P}(X \leq 35) = 0.0280 > 0.025$

$\mathrm{P}(X \geq 43) = 0.0571 > 0.025$

$\mathrm{P}(X \geq 44) = 0.0097 < 0.025$

CR is $\{0 \leq X \leq 34\} \cup \{X = 44\}$

Actual Significance Level = 2.00%

8 c)

$\mathrm{P}(X \leq 9) = 0.0334 < 0.05$

$\mathrm{P}(X \leq 10) = 0.0663 > 0.05$

$\mathrm{P}(X \geq 21) = 0.0658 > 0.05$

$\mathrm{P}(X \geq 22) = 0.0366 < 0.05$

CR is $\{0 \leq X \leq 9\} \cup \{22 \leq X \leq 55\}$

Actual Significance Level = 7.00%

8 d)

$\mathrm{P}(X \leq 3) = 0.0015 < 0.005$

$\mathrm{P}(X \leq 4) = 0.0052 > 0.005$

$\mathrm{P}(X \geq 21) = 0.0095 > 0.005$

$\mathrm{P}(X \geq 22) = 0.0047 < 0.005$

CR is $\{0 \leq X \leq 3\} \cup \{22 \leq X \leq 135\}$

Actual Significance Level = 0.611%

9 a)

Firstly, the alternative hypothesis should be H_1: $p < 0.25$ and not H_1: $p \leq 0.25$ (alternative hypothesis should be a strict inequality). Secondly, the p-value to be found should be $\mathrm{P}(X \leq 5)$ and not $\mathrm{P}(X = 5)$ (it should be a cumulative probability).

9 b)

Let X be the number customers purchasing a chrysanthemum bouquet.

$\mathrm{H}_0 : p = 0.25$

$\mathrm{H}_1 : p < 0.25$

$X \sim \mathrm{B}(34, 0.25)$

$\mathrm{P}(X \leq 5) = 0.1138$

$0.1138 > 0.1$ so we fail to reject H_0

There is insufficient evidence to suggest that the proportion of customers purchasing a chrysanthemum bouquet has decreased.

10)

Let X be the number of games Chaisai wins.

$\mathrm{H}_0 : p = \frac{1}{5}$

$\mathrm{H}_1 : p > \frac{1}{5}$

$X \sim \mathrm{B}(45, \frac{1}{5})$

$\mathrm{P}(X \geq 15) = 0.0250$

$0.0250 < 0.05$ so we reject H_0

There is sufficient evidence to suggest that the proportion of games won by Chaisai has increased.

11)

Let X be the number of adults who met the fruit recommendation of 2 servings a day.

$\mathrm{H}_0 : p = 0.441$

$\mathrm{H}_1 : p < 0.441$

$X \sim \mathrm{B}(60, 0.441)$

$\mathrm{P}(X \leq 20) = 0.0593$

$0.0593 > 0.05$ so we fail to reject H_0

There is insufficient evidence to suggest that the proportion of adults who met the fruit recommendation of 2 servings a day is less than 44.1%.

12)

Let X be the number of employed adults.

$\mathrm{H}_0 : p = 0.758$

$\mathrm{H}_1 : p < 0.758$

$X \sim \mathrm{B}(110, 0.758)$

$\mathrm{P}(X \leq 70) = 0.0029$

$0.0029 < 0.01$ so we reject H_0

There is sufficient evidence to suggest that the proportion of employed adults in Brighton is lower than the UK average.

13)

Let X be the number of batteries that last longer than 20 hours.

$\mathrm{H}_0 : p = 0.77$

$\mathrm{H}_1 : p > 0.77$

$X \sim \mathrm{B}(150, 0.77)$

$\mathrm{P}(X \geq 126) = 0.0229$

$0.0229 < 0.05$ so we reject H_0

There is sufficient evidence to suggest that the proportion of batteries lasting longer than 20 hours has increased.

14)

Let X be the number of TV remotes that have a fault.

$\mathrm{H}_0 : p = 0.04$

$\mathrm{H}_1 : p \neq 0.04$

$X \sim \mathrm{B}(60, 0.04)$

$60 \times 0.04 = 2.4$

$\mathrm{P}(X \geq 6) = 0.0325$

$0.0325 > 0.025$ so we fail reject H_0

There is insufficient evidence to suggest that the proportion of TV remotes that have a fault has changed.

15)

Let X be the number of roses that suffer from powdery mildew.

$\mathrm{H}_0 : p = 0.35$

$\mathrm{H}_1 : p \neq 0.35$

$X \sim \mathrm{B}(300, 0.35)$

$300 \times 0.35 = 105$

$\mathrm{P}(X \leq 81) = 0.0019$

$0.0019 < 0.015$ so we reject H_0

There is sufficient evidence to suggest that the proportion of roses that suffer from powdery mildew has changed.

16 a)

$\mathrm{H}_0 : p = \frac{1}{6}$

$\mathrm{H}_1 : p > \frac{1}{6}$

16 b)

Let X be the number of times the die rolls a 1.

$X \sim \mathrm{B}(35, \frac{1}{6})$

$\mathrm{P}(X \geq 9) = 0.1162 > 0.1$

$\mathrm{P}(X \geq 10) = 0.0550 < 0.1$

CR is $\{10 \leq X \leq 35\}$

16 c)

7 is not in the critical region, so we fail to reject H_0

There is insufficient evidence to suggest that the proportion of times the die rolls a 1 is greater than $\frac{1}{6}$.

17 a)

$H_0 : p = 0.76$

$H_1 : p < 0.76$

17 b)

Let X be the number of patients who must wait longer than 3 hours.

$X \sim B(70, 0.76)$

$P(X \leq 44) = 0.0096 < 0.01$

$P(X \leq 45) = 0.0185 > 0.01$

CR is $\{0 \leq X \leq 44\}$

17 c)

43 is in the critical region, so we reject H_0

There is sufficient evidence to suggest that the proportion of patients who wait longer than 3 hours has reduced.

18 a)

$H_0 : p = 0.32$

$H_1 : p \neq 0.32$

18 b)

Let X be the number of customers who bought packs of six of soup.

$X \sim B(80, 0.32)$

$P(X \leq 17) = 0.0232 < 0.025$

$P(X \leq 18) = 0.0413 > 0.025$

$P(X \geq 34) = 0.0312 > 0.025$

$P(X \geq 35) = 0.0182 < 0.025$

CR is $\{0 \leq X \leq 17\} \cup \{35 \leq X \leq 80\}$

18 c)

The actual significance level is $P(X \leq 17) + P(X \geq 35) = 0.02319241 + 0.01818237 = 0.0414$, so 4.14%

18 d)

45 is in the critical region, so we reject H_0

There is sufficient evidence to suggest that the proportion of customers purchasing single cans of soup has changed.

19 a)

$H_0 : p = 0.05$

$H_1 : p < 0.05$

19 b)

Let X be the number of bags of grain that are spoilt.

$X \sim B(550, 0.05)$

$P(X \leq 20) = 0.0807 < 0.1$

$P(X \leq 21) = 0.1175 > 0.1$

CR is $\{0 \leq X \leq 20\}$

19 c)

13 is in the critical region, so we reject H_0

There is sufficient evidence to suggest that the proportion of spoilt bags of grain from Baker Farms is less than that of Aaron Farms.

20 a i)

$Y \sim B(25, 0.08)$

$P(Y = 1) = 0.2704$

20 a ii)

$P(Y \leq 2) = 0.6768$

20 b i)

Let X be the number of gold dinosaurs.

$H_0 : p = 0.08$

$H_1 : p > 0.08$

20 b ii)

$X \sim B(30, 0.08)$

$P(X \geq 5) = 1 - P(X \leq 4) = 0.0874 > 0.05$

$P(X \geq 6) = 1 - P(X \leq 5) = 0.0293 < 0.05$

CR is $\{6 \leq X \leq 30\}$

20 b iii)

4 is not in the critical region, so we fail to reject H_0

There is insufficient evidence to suggest that the proportion of gold dinosaurs found is higher when shopping through the website.